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Roll No

EC-223-CBCS

B.E. III Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Electronic Devices and Circuits

Time : Three Hours

Maximum Marks : 60

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1.
 - a) Discuss about Drift current and diffusion current.
 - b) Explain the working of PN junction diode in forward and reverse bias.
2. Draw the symbol and explain the working and application of following diodes.
 - i) Zener diode
 - ii) Varactor diode
 - iii) Schottky diode
 - iv) LED
3.
 - a) Discuss the construction and basic operation of bipolar junction transistor.
 - b) Discuss about Eber's Moll model.
4.
 - a) Discuss about voltage divider bias and collector to base bias of transistor.
 - b) Discuss the working of transistor as a switch.

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5. Discuss the analysis of transistor amplifier using h-parameters.
6. Discuss the working of following power amplifiers.
 - i) Class A
 - ii) Class AB
7.
 - a) Discuss the principle working of enhancement and depletion MOSFET.
 - b) Discuss the principle of operation of UJT.
8. Write short notes on any two of the following.
 - a) Tunnel diode
 - b) Hall effect
 - c) Silicon controlled rectifier

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